# Shaikh Akib Shahriyar

Ph.D. Candidate in Computing and Information Sciences

akibshahriyar8751@gmail.com +1 (585) 309-8657 Rochester, NY
linkedin.com/in/akibshahriyar Google Scholar Homepage

*AI Security • Adversarial Machine Learning • Multimodal Robustness • Secure Agentic AI • Synthetic-Media Forensics*

## Education

**Ph.D. in Computing and Information Sciences** Spring 2027 (expected)
Rochester Institute of Technology, Rochester, NY
Advisor: Dr. Matthew Wright
*Dissertation: RobustDF: Developing a Practical and Robust Defense Framework for Deepfake Detection*

**M.Sc. Eng. in Computer Science and Engineering** 2019
Khulna University of Engineering & Technology, Khulna, Bangladesh
*Thesis: Speech Enhancement Using Convolutional Denoising Autoencoder*

**B.Sc. Eng. in Computer Science and Engineering** 2017
Khulna University of Engineering & Technology, Khulna, Bangladesh
*Thesis: An Efficient Error Correction Approach by Using Successive Parity Generation*
Ranked **1st of 59** in Computer Science and Engineering and **1st of 390** in the Faculty of Electrical and Electronic Engineering.

## Academic Appointments

**Adjunct Faculty** Dec. 2024 – May 2025
Rochester Institute of Technology *Rochester, NY*

**Assistant Professor** Jan. 2021 – Sep. 2023
Khulna University of Engineering & Technology (KUET) *Khulna, Bangladesh*
*Held concurrently with Ph.D. studies through approved institutional study leave until formal resignation in September 2023.*

**Lecturer** Jan. 2018 – Dec. 2020
Khulna University of Engineering & Technology (KUET) *Khulna, Bangladesh*

**Lecturer** May 2017 – Dec. 2017
North Western University (NWU) *Khulna, Bangladesh*

## Research Interests

AI security and Adversarial machine learning; Multimodal practical robustness; Deepfake and synthetic-media forensics; Multi-Agent system security; Task-preserving attacks and authorization-aware agent evaluation; Persistent-memory security.

## Under Review/Working Manuscripts

**U1.** **Shaikh Akib Shahriyar**, Ben Brown, Armon Barton, Ning Zhang, Matthew Wright, "SoK: The Practical Robustness Frontier: Cost-Aware Vulnerability Analysis for Deepfake Detectors," *Manuscript in preparation for the 48th IEEE Symposium on Security and Privacy (S&P)*, 2027.

**U2.** **Shaikh Akib Shahriyar**, Akil Iftee, Armon Barton, Matthew Wright, "What Counts as the Same Task? Executable Contracts for Robustness Evaluation of Tool-Using Agents," *under review at the 26th International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, 2027.

**U3.** **Shaikh Akib Shahriyar**, Zinedine Sehili, Adam Arthur, Christopher Schwartz, "From Augmentation to Synthetic Evidence: Evaluating Generated Chest Radiographs at Dataset Intake," *under review at SAFE workshop at the 41st Annual AAAI Conference on Artificial Intelligence (AAAI)*, 2027.

# Peer-Reviewed Publications

## Journal Articles

**J1.** Armon Barton, Matthew Wright, **Shaikh Akib Shahriyar**, Edgar Jatho, Mohammad Saidur Rahman, Kantha Girish Gangadhara, Jiang Ming, "PadNet: Defending Neural Networks Against Adversarial Examples," *ACM Transactions on Privacy and Security*, 2026.

**J2.** Anas Basalamah, Mahedi Hasan, Shovan Bhowmik, **Shaikh Akib Shahriyar**, "A Highly Accurate Dysphonia Detection System Using Linear Discriminant Analysis," *Computer Systems Science and Engineering*, 2023.

**J3.** M. A. H. Akhand, Safial Islam Ayon, **Shaikh Akib Shahriyar**, N. Siddique, H. Adeli, "Discrete Spider Monkey Optimization for Traveling Salesman Problem," *Applied Soft Computing*, 2020.

## Conference & Workshop Papers

**C1.** Anton Selitskiy, **Shaikh Akib Shahriyar**, Jishnuraj Prakasan, "Discrete Optimal Transport is a Strong Audio Adversarial Attack," *IEEE/IAPR International Joint Conference on Biometrics (IJCB)*, to appear, 2026.

**C2.** **Shaikh Akib Shahriyar**, Matthew Wright, Armon Barton, "IVORY: Adversarial Purification of Obfuscated Faces to Extract Soft-Biometrics using Diffusion Transformers," *IEEE International Conference on Automatic Face and Gesture Recognition (FG)*, 2025.

**C3.** **Shaikh Akib Shahriyar**, Matthew Wright, "Evaluating Robustness of Sequence-based Deepfake Detector Models by Adversarial Perturbation," *Workshop on Security Implications of Deepfakes and Cheapfakes (WDC @ ACM AsiaCCS)*, 2022.

**C4.** **Shaikh Akib Shahriyar**, Shovan Bhowmik, Mahedi Hasan, "DNA Cryptography-Based Speech Security System," *International e-Conference on Intelligent Systems and Signal Processing (eISSP)*, Springer, 2022.

**C5.** Iftekher Toufique Imam, Yamin Arafat, Kazi Saeed Alam, **Shaikh Akib Shahriyar**, "DOC-BLOCK: A Blockchain Based Authentication System for Digital Documents," *International Conference on Intelligent Communication Technologies and Virtual Mobile Networks (ICICV)*, IEEE, 2021.

**C6.** Raisa Tahsin, Mahmudul Hasan Mozumder, **Shaikh Akib Shahriyar**, Md Abdus Salim Mollah, "A Novel Approach for E-mail Classification Using FastText," *IEEE Region 10 Symposium (TENSYMP)*, 2020.

**C7.** M. I. R. Shuvo, **Shaikh Akib Shahriyar**, M. A. H. Akhand, "Bangla Numeral Recognition from Speech Signal Using Convolutional Neural Network," *International Conference on Bangla Speech and Language Processing (ICBSLP)*, IEEE, 2019.

**C8.** Safial Islam Ayon, M. A. H. Akhand, **Shaikh Akib Shahriyar**, N. Siddique, "Spider Monkey Optimization to Solve Traveling Salesman Problem," *International Conference on Electrical, Computer and Communication Engineering (ECCE)*, IEEE, 2019.

**C9.** **Shaikh Akib Shahriyar**, M. A. H. Akhand, N. Siddique, T. Shimamura, "Speech Enhancement Using Convolutional Denoising Autoencoder," *International Conference on Electrical, Computer and Communication Engineering (ECCE)*, IEEE, 2019.

**C10.** **Shaikh Akib Shahriyar**, K. M. Rokibul Alam, S. S. Roy, Y. Morimoto, "An Approach for Multi Label Image Classification Using Single Label Convolutional Neural Network," *International Conference of Computer and Information Technology (ICCIT)*, IEEE, 2018.

**C11.** **Shaikh Akib Shahriyar**, M. A. Golap, M. S. Sadi, "An Efficient Error Correction Approach by Using Successive Parity Generation," *International Conference on Electrical Engineering and Information & Communication Technology (iCEEiCT)*, IEEE, 2018.

**C12.** S. S. Roy, **Shaikh Akib Shahriyar**, M. Asaf-Uddowla, K. M. R. Alam, Y. Morimoto, "A Novel Encryption Model for Text Messages Using Delayed Chaotic Neural Network and DNA Cryptography," *International Conference of Computer and Information Technology (ICCIT)*, IEEE, 2017.

# Archival Posters

**P1.** **Shaikh Akib Shahriyar**, Ben Brown, Armon Barton, Ning Zhang, Matthew Wright, "How Many Queries Does It Take to Break a Deepfake Detector? Preliminary Findings from a Cost-Aware Robustness Audit," *35th USENIX Security Symposium*, 2026.

**P2.** **Shaikh Akib Shahriyar**, Matthew Wright, Armon Barton, "DiffTranPure: Diffusion Transformer-based Adversarial Purification to Extract Face and Facial Soft-Biometric Features from Obfuscated Images," *33rd USENIX Security Symposium*, 2024.

# Research Experience

**Graduate Research Assistant** Aug. 2020 – Present
ESL Global Cybersecurity Institute, Rochester Institute of Technology *Rochester, NY*
*Advisor: Dr. Matthew Wright*

- Developed **RobustDF** and the **Practical Robustness Frontier (PRF)**, a cost-aware evaluation framework relating attack success to adversary access, query budgets, transferability, and preserved utility across 12 deepfake detectors from four architectural families (**U1, P1, C3**).
- Developed an executable **task-contract framework** that separates task identity from evaluation admissibility, using representation proxies and fixed-trajectory rescoring across 14,336 trajectories to isolate how membership rules alter agent-security conclusions (**U2**).
- Designed and evaluated black-box optimal-transport attacks on audio anti-spoofing and Diffusion Transformer purification (**IVORY**) for obfuscated facial soft-biometrics (**C1, C2, P2**).
- Contributed to **PadNet**'s defensive barrier-class architecture, gradient regularization, and adaptive evaluation against rejection-targeted evasion (**J1**).
- Collaborated across RIT, the Naval Postgraduate School, and partner institutions on adversarial ML, synthetic-media forensics, and secure agent evaluation (**U1, U3, J1**).

# Sponsored Research Contributions

**VARUNA: A User-Centered All-in-One Media Forensics Platform** 2025 – Present
**Research Contributor**; National Science Foundation Award 2429835; **$1.5M total award** ($1.125M to RIT).
Contribute adversarial-robustness and deepfake-detection research to a multi-institutional platform for journalists and forensic analysts.

**Deepfake Detection for Journalists / DeFake Project** 2020 – Present
**Research Contributor**; supported by the National Science Foundation and the John S. and James L. Knight Foundation, including a **$200,000 Knight Foundation award**.
Developed adversarial evaluations and robustness methods for usable, interpretable media-forensics tools within a program with more than **$2M** in combined support.

# Teaching Experience

**Instructor of Record, CSEC 720: Deep Learning Security** Spring 2025
Rochester Institute of Technology *Rochester, NY*
*Graduate Seminar (13 M.S. and junior Ph.D. students)*

- Designed and delivered complete graduate curriculum bridging modern deep learning with adversarial robustness, privacy, and robust evaluation.
- Created hands-on coding laboratories and open-ended research projects resulting in conference-quality student work.

**Instructor of Record, CSEC 140: Introduction to Cybersecurity** Fall 2022, Fall 2026
Rochester Institute of Technology *Rochester, NY*
*Undergraduate Core Course ( 30 students)*

- Taught fundamental principles of threat modeling, cryptography, network defense, systems security, and privacy to undergraduate computing majors.

**Assistant Professor & Lecturer** Jan. 2018 – Sep. 2023
Khulna University of Engineering & Technology (KUET) *Khulna, Bangladesh*
*Department of Computer Science and Engineering*

- Served as primary instructor for core undergraduate and graduate courses including *Machine Learning, Computer Security, Mobile Computing, Computer Networks, Software Engineering,* and *Algorithm Analysis & Design.*
- Managed large enrollments ( 120 students across multiple sections per course), delivering lectures, supervising laboratory sessions, designing exams, and mentoring undergraduate capstone projects.

**Lecturer** May 2017 – Dec. 2017
North Western University (NWU) *Khulna, Bangladesh*
*Department of Computer Science and Engineering*

- Taught foundational courses (*Artificial Intelligence, Software Engineering, Computer Networks*) to cohorts of 40 students.

**Instructor (Professional Development)** 2018 – 2020
Govt. Teachers' Training Institute *Khulna, Bangladesh*
*K–12 Educator Training Program*

- Taught specialized development modules in *Machine Learning* and *Computer Networks* to 20 K–12 educators per cohort.

# Student Mentoring

**Ph.D. Student(s)**

- Ben Brown, Ph.D. Student in Electrical and Computer Engineering, RIT.
- Y. Kelly Wu, Ph.D. Student in Computing and Information Sciences, RIT.
- Bryce Gernon, Ph.D. Student in Computing and Information Sciences, RIT.
- Ishmam Bin Rofi, Ph.D. Student in Computing and Information Sciences, RIT.
- Akil Iftee, Ph.D. Student in Computing and Information Sciences, RIT.

**BS/MS Student(s)**

- Zinedine Sehili, BS in Computer Science, RIT.
- Adam Arthur, BS in Cybersecurity, RIT.
- Rajat Sahay, MS in Data Science, RIT.
- Anurag Mishra, MS in Data Science, RIT.
- Matt Emerick, BS/MS in Cybersecurity, RIT.
- Robert McCall, BS/MS in Cybersecurity, RIT.
- Trevor Jarrett, BS in Cybersecurity, RIT.
- Christopher Sorrentino, BS in Cybersecurity, RIT.
- Oleksandr Maniukhin, BS in Cybersecurity, RIT.
- Shane Russell, BS in Cybersecurity, RIT.
- Meghan Briskey, BS in Cybersecurity, RIT.
- Jonathan Ho, BS in Cybersecurity, RIT.
- Shree Kumar, BS in Cybersecurity, RIT.
- Aidan Morgan, BS in Cybersecurity, RIT.
- Alexander Kilroy, BS in Cybersecurity, RIT.
- Griffin Vella, BS in Cybersecurity, RIT.
- Alexander Noble, BS in Cybersecurity, RIT.
- Jeffrey Burke, BS in Cybersecurity, RIT.
- Evelyn Gerber, BS in Cybersecurity, RIT.
- Jared Korwatch, BS in Cybersecurity, RIT.
- Anushka Tilara, BS in Cybersecurity, RIT.

**Undergraduate Research and Capstone Advising** 2017 – 2023
Khulna University of Engineering & Technology *Khulna, Bangladesh*
Mentored more than 30 students and co-authored 7 student-led peer-reviewed publications in machine learning, computer vision, security, speech processing and blockchain systems.

# Technical Leadership & Venture Experience

**Principal ML Architecture Consultant (Part-time)** Sep. 2023 – Sep. 2025
Apprendi AI *Stanford, CA*

- Designed the core research-to-product architecture for a multimodal agentic system, combining LLMs & diffusion models, establishing the technical infrastructure, tool orchestration, and evaluation guardrails from the ground up.
- Scaled deployment to support user-facing workflows; adopted by 95 child psychiatrists and psychologists across 14 schools in four districts.

**Technical Strategy Advisor & Co-Investor** Aug. 2023 – Sep. 2025
Zatiq *Dhaka, Bangladesh*

- Provided strategic technical oversight for an emerging-market fintech platform focused on SME e-commerce democratization.
- Advised on scalable data infrastructure and technical product roadmap for regional deployment.

**Lead Machine Learning Research Engineer** Jun. 2019 – Dec. 2020
SmartRoad.AI - Remote *Vitry-sur-Seine, France*

- Led a four-engineer team developing automatic number-plate recognition and manipulated-plate detection; achieved 98% detection accuracy and reduced inference time by 47% for edge deployment.

# Awards and Honors

**Prime Minister Gold Medal of Bangladesh**; ranked first in the Faculty of Electrical and Electronic Engineering, KUET. 2017

**President Gold Medal of Bangladesh**; ranked first in the Department of Computer Science and Engineering, KUET. 2017

**Best Student Poster Award**, UPSTAT, Rochester, NY. 2023

**Best Student Poster Award**, IEEE Western New York Image and Signal Processing Workshop. 2021

# Selected Talks and Presentations

| | |
|---|---|
| Sep. 2026 | **When Accuracy Is Not Security: Measuring and Improving Practical Robustness in Deepfake Detection.** GCCIS PhD Colloquium Series, Rochester Institute of Technology, Rochester, NY. |
| Aug. 2026 | **How Many Queries Does It Take to Break a Deepfake Detector? Preliminary Findings from a Cost-Aware Robustness Audit.** Poster, 35th USENIX Security Symposium, Baltimore, MD. |
| May 2025 | **Towards Trustworthy Deepfake Detection: Systematizing Adversarial Robustness and Charting Future Defenses.** Doctoral Consortium, IEEE FG, Clearwater, FL. |
| Aug. 2024 | **DiffTranPure: Diffusion Transformer-based Adversarial Purification to Extract Face and Facial Soft-Biometric Features from Obfuscated Images.** Poster, 33rd USENIX Security Symposium, Philadelphia, PA. |
| Jan. 2023 | **Adversarial ML to Adversarial Deepfakes: Attacks and Countermeasures.** Guest talk, CSEC 720 Deep Learning Security, Rochester Institute of Technology. |

# Professional Service

| | |
|---|---|
| **Program Committee (PC)** | International Conference on Autonomous Agents and Multiagent Systems (AAMAS) 2027 |
| **Program Committee (PC)** | IEEE International Conference on Automatic Face and Gesture Recognition (FG) 2026, 2027 |
| **Program Committee (PC)** | EAI International Conference on Digital Forensics and Cyber Crime (ICDF2C) 2020–2023 |
| **Journal Reviewer** | ACM Transactions on Privacy and Security (TOPS)<br>IEEE Transactions on Dependable and Secure Computing (TDSC)<br>IEEE Transactions on Information Forensics and Security (TIFS)<br>IEEE Access<br>International Journal of Electronics and Communications (IJECE) |
| **Conference Reviewer** | AAMAS 2027<br>IEEE FG 2025–2027<br>IEEE/CVF CVPR 2026<br>EAI ICDF2C 2020–2022<br>IEEE TCCE 2020 |
| **Academic Leadership** | **Vice President, Doctoral Student Association, RIT**, 2022–2025<br>Coordinated professional-development programming and conference-participation awards for Ph.D. students. |